

Tamagno Roy

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Education

- **RCC Institute of Information Technology (RCCIT)**

Bachelor in Technology (Computer Science and Engineering (AI&ML)); CGPA: 8.05

Kolkata, India
July 2023 – Present
- **Douglas Memorial Higher Secondary School (CISCE)**

Higher Secondary (stream: PCMB); Percentage: 87%

Barrackpore, India
Apr 2021 – Apr 2023

Experience

- **Jawd Life Styles (Intern)**

Kolkata, India
Apr 2024 – July 2024

 - **Developed an internal bulk mailing system** using Next.js, Firebase and Node.js following production grade software engineering practices.
 - **Implemented Server-Sent Events (SSE)** for real-time logging, gaining hands-on experience with streaming data, performance considerations, and debugging asynchronous systems. Led integration efforts, optimizing system workflows and improving efficiency by 15% .
 - **Developed reusable frontend components** and authored technical documentation to improve maintainability and team collaboration.
 - **Authored clear technical documentation and reusable components**, ensuring long-term maintainability and smoother collaboration across the development team.
 - **Tech Stack:** TypeScript, Next.js, Firebase, Node.js, Express.js, Tailwind CSS, REST APIs.
- **Summer Research Intern (NPTEL Summer Internship Program 2026)**

Indian Institute of Technology (IIT) Ropar

 - Working on software engineering and AI-driven projects while gaining hands-on experience in modern development practices and collaborative problem solving.
 - Participating in an 8-week virtual internship program focused on technical skill development and project implementation.

TECHNICAL SKILLS

- **Languages:** JavaScript, TypeScript, HTML5, CSS3, Python, C, Java, SQL
- **Frontend:** React.js, Next.js, Redux, Tailwind CSS
- **Backend:** Node.js, Express.js, FastAPI, REST APIs
- **Artificial Intelligence:** LangChain, LangGraph, Hugging Face, TensorFlow (Basics)
- **Developer Tools:** Git, GitHub, Postman, VS Code, Docker
- **Database:** MongoDB, MySQL, SQLite, Postgres.

Projects

- **EduAgents | AI-Powered Adaptive Learning Platform**

Next.js, TypeScript, Supabase, Vercel AI SDK, LangChain, Hume AI

Frontend + AI integration
 - Built an AI-powered adaptive learning platform that dynamically generates personalized curricula, assessments, and educational resources using Google Gemini and agentic AI workflows.
 - Developed real-time conversational tutoring with voice-enabled interactions, speech-to-text processing, and emotion-aware responses using Hume AI Voice SDK.
 - Implemented scalable backend services with Supabase (PostgreSQL) and Vercel AI SDK to support streaming LLM responses, session management, and progress tracking.
 - Designed adaptive assessments and AI-generated assignments with performance-based difficulty adjustment and instant feedback mechanisms.
 - Links: [Github repository.](#)
- **Gradify.AI – AI-Based Handwritten Answer Evaluation System**

Python, OCR, LLMs, Semantic Evaluation, REST APIs, Postman, Git

Backend+ AI integration
 - **Built an end-to-end AI system to automatically evaluate handwritten assignments**, using OCR for text extraction and LLM-based answer generation to compare student responses against model solutions.
 - **Designed scalable REST APIs** to extract handwritten text, generate context-aware model answers from uploaded study materials, and compute marks based on semantic similarity and assigned weightage.
 - **Reduced manual grading effort for educators** by enabling bulk evaluation, real-time scoring, and standardized assessment through an automated, data-driven pipeline.
 - LINKS:- [Github Repository.](#)

Honors / Achievements

- Hack <N> Pitch 1ST Runner Up (Agentic AI) : Secured 2nd position in this hackathon organized by **Jadavpur University**.
- **Web Development Lead**, Hult Prize RCCIT
- **Deputy Campus Director**, Hult Prize RCCIT (2025-26)
- Selected for the **NPTEL Summer Internship Program 2026 at IIT Ropar** from a nationwide applicant pool for an 8-week research internship.